

Traffic Rule Summary

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1 Introduction

This document serves as an overview about formalized traffic rules based on the German Road Traffic Regulation (Straßenverkehrs-Ordnung), in the following abbreviated with StVO, and the Vienna Convention on Road Traffic [1], in the following abbreviated with VCoRT.

The traffic rules formalization process is separated in four steps:

1. **Extracting rules from legal sources:** The rules of the road are specified in the traffic laws of a country or region. However, often traffic rules are not precise and concrete. Therefore, judicial decisions which interpret the traffic laws have to be considered. Additionally, it is often the case that several laws have to be considered at the same time because the rules interrelate. Please note that no official translated version of the StVO exist. The translation used in this document is based on the *German Law Archive* website¹.
2. **Concretizing traffic rules and legal judicial decisions:** The considered traffic rules and judicial decisions have to be combined to a sentence which on the one side describes the situation where the rule is applicable and on the other side concretizes the different legal sources. It can also be the case that a single traffic rule addresses different behaviors and therefore have to be divided (e.g., StVO §4 (1)).
3. **Extracting predicates:** For the logical formalization of traffic rules one needs predicates for the evaluation of certain properties, e.g., allowed speed limit or velocity of preceeding vehicles.
4. **Defining temporal logic formulas:** The last step is to formalize the concretized sentence using the extracted predicates in temporal logic.

We formalize the traffic rules in Linear Temporal Logic (LTL), Metric Temporal Logic (MTL), and Signal Temporal Logic (STL).

The remainder of this paper is structured as follows. Sec. 2 introduces mathematical foundations and different temporal logics. Sec. 4 presents the formalization of traffic rules. The traffic rules are clustered in general traffic rules (Sec. 4.2) applicable in all possible traffic situations, highway traffic rules (Sec. 4.3), urban traffic rules (Sec. 4.4), and cross-country road rules (Sec. 4.5). In Sec. 3 predicates for the temporal logic formulas are defined.

¹<https://germanlawarchive.iuscomp.org/?p=1290>

2 Preliminaries

2.1 Road Network

We describe our road network with lanelets, which are atomic, drivable road segments, geometrically represented by a left and right bound, where the bounds are represented by polylines [2]. The set of all lanelets is denoted by $\mathbb{L}$. In the following we refer to a single lanelet as l , where $l \in \mathbb{L}$. The functions $l_{lb}(l)$, $l_{rb}(l)$, $l_{begin}(l)$, and $l_{end}(l)$ return the left bound, right bound, left and right beginning points, and left and right ending points of a lanelet. The functions

$$adj_l(l) = \begin{cases} l' & \text{if } \exists l' \in \mathbb{L} : l_{lb}(l') = l_{rb}(l) \\ \emptyset & \text{otherwise} \end{cases} \quad (1)$$

$$adj_r(l) = \begin{cases} l' & \text{if } \exists l' \in \mathbb{L} : l_{rb}(l) = l_{lb}(l') \\ \emptyset & \text{otherwise} \end{cases} \quad (2)$$

$$succ(l) = \begin{cases} l' & \text{if } \exists l' \in \mathbb{L} : l_{end}(l) = l_{begin}(l') \\ \emptyset & \text{otherwise} \end{cases} \quad (3)$$

$$pre(l) = \begin{cases} l' & \text{if } \exists l' \in \mathbb{L} : l_{begin}(l) = l_{end}(l') \\ \emptyset & \text{otherwise} \end{cases} \quad (4)$$

return the adjacent left lanelet, adjacent right lanelet, successor lanelets, and predecessor lanelets of l , respectively. The curvature of a lanelet at a given path length s_l is given by $curv(s_l)$. The functions $speed_limit()$ and $required_speed()$ return the maximum and the minimum required speed which a vehicle must consider on a lanelet. This information is modeled by traffic signs which belong to the lanelet. Additionally, we assume that an environmental analysis exists which provides a maximum velocity Δv_H which is allowed to drive based on the road conditions, e.g., due to snow or ice. A lanelet can have several attributes which further describe it. The set of all attributes is $\mathbb{A}$. For the highway case, we have $\mathbb{A} = \{main_carriage_way, access_ramp, exit_ramp, shoulder\}$. The function $attr(l)$ returns the attributes which are valid for a lanelet. Similar, the left and right bounds of a lanelet can have lanelet markings, with the set of lanelet markings $\mathbb{LM} = \{solid, dashed, broad_solid, broad_dashed\}$. The functions $mark_l(l)$ and $mark_r(l)$ return the left and right lanelet marking, respectively. We define the lane part in front of a lanelet, in the rear of a lanelet as follows:

$$lane_{front}(l) = \{l' \mid \forall l'', l''' \in (l' \in (succ(l) \cup lane_{front}(succ(l)))) : attr(l'') = attr(l''')\}, \quad (5)$$

$$lane_{rear}(l) = \{l' \mid \forall l'', l''' \in (l' \in (pre(l) \cup lane_{rear}(pre(l)))) : attr(l'') = attr(l''')\}. \quad (6)$$

We define the lane to which a lanelet belongs to, using the lanelets in front and rear of another lanelet:

$$lane(l) = \{l' \in (l \cup lane_{rear}(l) \cup lane_{front}(l)) \mid \forall l'', l''' \in l' : attr(l'') = attr(l''') \\ \wedge adj_l(l') \in succ((adj_l(pre(l')))) \wedge adj_r(l') \in succ((adj_r(pre(l'))))\}. \quad (7)$$

The left and right lane are defined as follows:

$$lane_{left}(l) = \{l' \mid \exists l'' \in lane(l) : l' = adj_l(l'')\}, \quad (8)$$

$$\text{lane}_{\text{right}}(l) = \{l' \mid \exists l'' \in \text{lane}(l) : l' = \text{adj}_r(l'')\}. \quad (9)$$

The functions

$$\text{lanes}_{\text{left}}(l) = \{l' \mid \exists l'' \in \text{lane}(l) : l' = \text{adj}_l(l'') \vee \exists l''' \in \text{lanes}_{\text{left}}(l'') : l' = \text{adj}_l(l''')\}, \quad (10)$$

$$\text{lanes}_{\text{right}}(l) = \{l' \mid \exists l'' \in \text{lane}(l) : l' = \text{adj}_r(l'') \vee \exists l''' \in \text{lanes}_{\text{right}}(l'') : l' = \text{adj}_r(l''')\} \quad (11)$$

return all left and right lanes of a given lanelet.

2.2 Vehicle

A vehicle V_i , $i \in \mathbb{N}^+$, consists of a longitudinal, lateral, and signal state $x_{\text{lon}} \in \mathbb{R}^n$, $x_{\text{lat}} \in \mathbb{R}^n$, and $x_{\text{sig}} \subseteq \text{SIG}$, respectively, where $\text{SIG} = \{\text{horn}, \text{indicator_left}, \text{indicator_right}, \text{braking_lights}, \text{hazard_warning_lights}, \text{flashing_blue_lights}\}$ represents possible active signals of a vehicle. The set of all vehicles is denoted by $\mathbb{V}$. The longitudinal state $x_{\text{lon}} = [s \ v \ a \ j]^T$ consists of position s , velocity v , acceleration a , and jerk j . The lateral state $x_{\text{lat}} = [d \ \theta \ \kappa \ \dot{\kappa}]^T$ consists of the lateral distance to the reference path Γ , orientation θ , curvature κ , and change of curvature $\dot{\kappa}$. The longitudinal input is $u(t) = \ddot{a}(t)$ and the lateral input is $u(t) = \ddot{\kappa}(t)$. The signal state contains only the subset of currently active elements of SIG . The occupancy of a vehicle for given states x_{lon} and x_{lat} is denoted by $\mathcal{O}(x_{\text{lon}}(t), x_{\text{lat}}(t))$. The function $\text{proj}()$ projects a vehicle to a position s . The functions $\text{front}()$ and $\text{rear}()$ return the front and rear position of a vehicle. All variables associated to the ego vehicle are denoted by the subscript $\square_{\text{ego}}$ and all variables associated to other vehicles are denoted by the subscript $\square_o$. For a given initial state, an input trajectory $u(\cdot)$, we introduce the solution of the model $\dot{x} = f(x, u)$ of the ego vehicle over time t as $\xi(t; x_0, u(\cdot))$. We denote a braking input by $u_{\text{brake}}(\cdot)$. The time at which a safe state is reached for $u(\cdot)$, given the system dynamics and x_0 , is denoted by $t_s(u(\cdot))$; we consider a state to be safe when one can stay in it indefinitely without causing a collision [3].

2.3 Linear Temporal Logic

2.4 Metric Temporal Logic

2.5 Signal Temporal Logic

3 Predicates

3.1 Position

$$\text{occupied_lanelets}(\mathbb{L}_i, V_i) = \{l' \mid l' \in \mathbb{L}_i : l' \cap \text{proj}(V_i) \neq \emptyset\} \quad (12)$$

$$\text{is_on_same_lane_as_ego_vehicle} \iff \text{occupied_lanelets}(\mathbb{L}_i, V_{\text{ego}}) \cap \text{occupied_lanelets}(\mathbb{L}_i, V_o) \neq \emptyset \quad (13)$$

$$\text{is_behind_ego_vehicle} \iff \text{front}(V_o) \leq \text{rear}(V_{\text{ego}}) \quad (14)$$

$$\text{is_in_front_of_ego_vehicle} \iff \text{front}(V_{\text{ego}}) \leq \text{rear}(V_o) \quad (15)$$

$$\text{is_in_front_of_ego_vehicle} \iff \text{front}(V_{\text{ego}}) \leq \text{rear}(V_o) \quad (16)$$

$$\begin{aligned} \text{vehicles_in_front_of_ego_vehicle} &= \{V' \mid V' \in \mathbb{V} : \text{occupied_lanelets}(\mathbb{L}, V') \\ &\subseteq \text{lane_front}(\text{occupied_lanelets}(\mathbb{L}, V_{\text{ego}})) \cup \text{occupied_lanelets}(\mathbb{L}, V_{\text{ego}})\} \end{aligned} \quad (17)$$

3.2 Velocity

Speed limit provided by traffic regulations for a vehicle V_i :

$$v_{\text{sl}}^{\max} := \sup (\{v \mid \forall l \in \text{occupied_lanelets}(\mathbb{L}, V_i) : v = \text{speed_limit}(l)\}) \quad (18)$$

Required speed limit provided by traffic regulations for a vehicle V_i :

$$v_{\text{sl}}^{\min} := \sup (\{v \mid \forall l \in \text{occupied_lanelets}(\mathbb{L}, V_i) : v = \text{required_speed}(l)\}) \quad (19)$$

The ego vehicle must always be able to stop within the field of view. Therefore, the maximum allowed velocity based on the field of view v_{sr} is chosen such that

$$\begin{aligned} v_{\text{fov}} &:= \text{proj}(\xi(t_s(\tilde{u}_{\text{brake}}(\cdot)); x_{\text{lon}}^{\text{sr}}, \tilde{u}_{\text{brake}}(\cdot))) \leq fov, \\ x_{\text{lon}}^{\text{sr}} &= [0, v_{\text{fov}}, \bar{a}, \bar{j}]^T. \end{aligned} \quad (20)$$

The ego vehicle must always be able reduce its velocity to a upcoming speed limit such that it does not brake abruptly. Therefore, the maximum allowed velocity must be limited such that the braking trajectory does not exceed acceleration and jerk thresholds.

$$\begin{aligned} v_{\text{br}} &:= \sup (\{v \mid x_{\text{f}} = \xi(t^*; x_0, u(\cdot)) \wedge \forall t \in [t_0, t^*] : a_{\text{ego}}(t) \geq a^* \wedge j_{\text{ego}}(t) \geq j^*\}) \\ x_0 &= [0, v_{\text{br}}, a_{\text{max,ego}}, j_{\text{max,ego}}]^T. \\ x_{\text{f}} &= [fov, v_{\text{sl}}^*, 0, 0]^T, \end{aligned} \quad (21)$$

where v_{sl}^* is the maximum speed limit which is allowed after free driving, e.g., in Germany if there is now active speed limit, the next speed limit is maximal 100 km/h. The velocity limits v_{fov} and v_{br} can be calculated offline using binary search. The field of view based velocity limit v_{fov} can change during runtime, e.g., due to changing weather conditions.

$$\text{keeps_lane_speed_limit} \iff v_{ego} \leq v_{sl}^{\max} \quad (22)$$

$$\text{keeps_fov_speed_limit} \iff v_{ego} \leq v_{fov} \quad (23)$$

$$\text{keeps_braking_speed_limit} \iff v_{ego} \leq v_{br} \quad (24)$$

$$\text{keeps_road_condition_speed_limit} \iff v_{ego} \leq v_w \quad (25)$$

$$\text{keeps_min_speed_limit} \iff v_{sl}^{\min} < v_{fov} \wedge v_{sl}^{\min} < v_w \implies v_{sl}^{\min} \leq v_{ego} \quad (26)$$

$$\text{preserves_traffic_flow} \iff \neg \text{slow_leading_vehicle} \implies v_{ego} - \text{vel_lead}() > \Delta v_{fl} \quad (27)$$

3.3 Safe distance

$$d_{\text{safe}}(v_{\text{rear}}, v_{\text{lead}}) = \begin{cases} d_{\text{safe},1}(v_{\text{rear}}, v_{\text{lead}}), & (a_{\text{min},\text{lead}} > a_{\text{min},\text{rear}}) \wedge (v_{\text{lead}}^* < v_{\text{rear}}) \wedge \\ & (t_d \leq \frac{v_{\text{lead}}}{-a_{\text{min},\text{lead}}}) \wedge (\frac{v_{\text{rear}}}{a_{\text{min},\text{rear}}} > \frac{v_{\text{lead}}^*}{a_{\text{min},\text{lead}}}), \\ d_{\text{safe},2}(v_{\text{rear}}, v_{\text{lead}}), & \text{otherwise,} \end{cases}$$

where

$$v_{\text{lead}}^* = \begin{cases} v_{\text{lead}} - a_{\text{min},\text{lead}} t_d, & t_d \leq \frac{v_{\text{lead}}(t_k)}{-a_{\text{min},\text{lead}}}, \\ 0, & \text{otherwise,} \end{cases}$$

$$d_{\text{safe},1}(v_{\text{rear}}, v_{\text{lead}}) := \frac{(v_{\text{lead}} - a_{\text{min},\text{lead}} t_d - v_{\text{rear}}^2)^2}{2(a_{\text{min},\text{lead}} - a_{\text{min},\text{rear}})} + v_{\text{rear}} t_d - v_{\text{lead}} t_d - \frac{1}{2} a_{\text{min},\text{lead}} t_d^2,$$

$$d_{\text{safe},2}(v_{\text{rear}}, v_{\text{lead}}) := \frac{v_{\text{lead}}^2}{2a_{\text{min},\text{lead}}} - \frac{v_{\text{rear}}^2}{2a_{\text{min},\text{rear}}} + v_{\text{rear}} t_d,$$

and t_d denotes the reaction time of the ego vehicle.

$$\text{keeps_safe_distance_front} \iff d_{\text{safe}}(v_{ego}, v_o) \leq (\text{rear}(V_o) - \text{front}(V_{ego})) \quad (28)$$

$$\text{keeps_safe_distance_rear} \iff d_{\text{safe}}(v_o, v_{ego}) \leq (\text{rear}(V_{ego}) - \text{front}(V_o)) \quad (29)$$

3.4 Overtaking

3.5 Unnecessary braking

$$\begin{aligned} \text{unnecessary_braking} &\iff (\text{velocity_reduction_necessary} \wedge a_{\text{ego}} < 0) \implies \\ &j_{\text{ego}} < j_{\text{abrupt}} \wedge \nexists V' \in \mathbb{V} : \text{vehicles_in_front_of_ego_vehicle} = V' \\ &\vee \exists V' \in \mathbb{V} : \text{vehicles_in_front_of_ego_vehicle} = V' \wedge (a_{\text{ego}} - a_0) < a_{\text{abrupt}} \wedge j_{\text{ego}} < j_{\text{abrupt}} \end{aligned} \quad (30)$$

4 Traffic Rule Formalization

Rules are formalized to account for general situations, e.g. the handling after collisions is not formalized.

4.1 Traffic Law Terminology

- access ramp / acceleration lane
- off ramp / deceleration lane
- main carriage way
- motor road / motor street / highway
- ego vehicle
- other vehicle

4.2 General Traffic Rules

4.2.1 Position

Safe distance front	
StVO §4 (1)	Der Abstand zu einem vorausfahrenden Fahrzeug muss in der Regel so groß sein, dass auch dann hinter diesem gehalten werden kann, wenn es plötzlich gebremst wird. A person operating a vehicle moving behind another vehicle must, as a rule, keep a sufficient distance from that other vehicle so as to be able to pull up safely even if it suddenly slows down or stops.
VCoRT §13 (5)	The driver of a vehicle moving behind another vehicle shall keep at a sufficient distance from that other vehicle to avoid collision if the vehicle in front should suddenly slow down or stop.
Concretization	A vehicle moving behind other vehicles within a lane must keep a distance which ensures collision freedom to each of these other vehicles even if one or several of them are suddenly slow down or stop.
Comments	Evaluated for all vehicles for which the following condition is valid: $\text{is_on_same_lane_as_ego_vehicle} \wedge \text{is_in_front_of_ego_vehicle}$
LTL	$G(\text{keeps_safe_distance_front})$
MTL	$G(\text{keeps_safe_distance_front})$
STL	$G(d_{\text{safe}}(v_{\text{ego}}, v_o) \leq (\text{rear}(V_o) - \text{front}(V_{\text{ego}})))$

Lane following vertical position	
StVO §2 (2)	Es ist möglichst weit rechts zu fahren, nicht nur bei Gegenverkehr, beim Überholtwerden, an Kuppen, in Kurven oder bei Unübersichtlichkeit. Drivers and riders must always keep as far to the right as possible, not only when traffic is approaching from the opposite direction, when being overtaken, when approaching the brow of a hill, on bends or when their view ahead is obstructed.
VCoRT §12 (1)	When passing oncoming traffic, a driver shall leave sufficient lateral space and, if necessary, move close to the edge of the carriageway appropriate to the direction of traffic. If in so doing he finds his progress impeded by an obstruction or by the presence of other road-users, he shall slow down and if necessary stop to allow the oncoming road-user or road-users to pass.
Concretization	If there exist only one lane on a road and vehicles can drive in both directions on this lane, the ego vehicle should drive as to the right as possible. If there exist at least one lane for each driving direction and the ego vehicle's lane is at least as broad as the ego vehicle or there is no adjacent lane in opposite driving direction, the ego vehicle should drive in the lane center.
LTL	$G((\text{only_one_lane} \wedge \text{no_lane_in_opposite_direction} \implies \text{drives_far_right_in_lane}) \wedge (\text{two_lanes_with_different_directions} \wedge \text{ego_lane_broad_enough_directions} \implies \text{drives_on_lane_center}))$
MTL	$G(\text{keeps_safe_distance_front})$
STL	$G(d_{\text{safe}}(v_{\text{ego}}, v_o) \leq (\text{rear}(V_o) - \text{front}(V_{\text{ego}})))$

4.2.2 Braking

Unnecessary braking	
StVO §4 (1)	Wer vorausfährt, darf nicht ohne zwingenden Grund stark bremsen. The person operating the vehicle in front must not brake suddenly without a compelling reason.
VCoRT §17 (1)	No driver of a vehicle shall brake abruptly unless it is necessary to do so for safety reasons.
Concretization	A vehicle shall never brake abruptly unless it has a reason to do so. A vehicle brakes abruptly if: 1. The vehicle brakes although there is no leading vehicle and there is no reason caused by a traffic regulation to reduce the vehicle's velocity; 2. There is an obstacle in front of the vehicle and the acceleration difference and jerk difference between the vehicle and the obstacle violate thresholds; 3. The maximum allowed speed will change or changed due to traffic signs or environmental conditions.
LTL	G(unnecessary_braking)
MTL	G(unnecessary_braking)
STL	
No stopping	
StVO §18 (8), §12 (1), OLG Karlsruhe DAR 02, 34, OLG Frankfurt DAR 01, 504, OLG Hamm VersR 91, 83, Bay VRS 59, 54 = StVE 23	Halten, auch auf Seitenstreifen, ist verboten. Das Halten ist unzulässig auf Einfädelungs- und auf Ausfädelungsstreifen. Das Anhalten auf allen Teilen der Autobahn, die dem Zweck dienen einen flüssigen Verkehr zu gewährleisten, wie die Autobahnen selbst, Zufahrtswege oder dem Seitenstreifen, ist verboten.
StVO English	
VCoRT §17 (1)	
Concretization	Stopping on the main carriage way, shoulder lane, and ramps is forbidden.
LTL	$G(\neg \text{in_congestion} \wedge (\text{on_access_ramp} \vee \text{on_exit_ramp} \vee \text{on_main_carriage_way} \vee \text{on_shoulder_lane}) \implies \neg \text{in_standstill})$
MTL	$G(\neg \text{in_congestion} \wedge (\text{on_access_ramp} \vee \text{on_exit_ramp} \vee \text{on_main_carriage_way} \vee \text{on_shoulder_lane}) \implies \neg \text{in_standstill})$
STL	

4.2.3 Velocity

Maximum Speed Limit	
StVO §3 (1), §3 (3), §18 (1), §18 (5), §18 (6)	The legal text is omitted for reasons of space.
VCoRT §13 (2)	Domestic legislation shall establish maximum speed limits for all roads. Domestic legislation shall also determine special speed limits applicable to certain categories of vehicles presenting a special danger, in particular by reason of their mass or their load. They may establish similar provisions for certain categories of drivers, in particular for new drivers.
Concretization	A vehicle shall not exceed the allowed speed limit of the lanes it drives on, shall not exceed the speed limit which is necessary to ensure collision freedom with respect to vehicles outside the field of view, and shall not exceed as speed which is necessary to comfortably ensure traffic regulations.
Comments	The traffic laws do not provide concrete information about speed limit, e.g., the VCoRT passes the task of defining speed limits to national lawmakers. The concretization assumes that a mapping of speed limits provided by traffic signs to lanes is given. If no speed limit is given, the speed limit caused by the field of view (traffic sign based speed limit set to infinity) and a speed limit which ensures comfortable braking for traffic regulations (e.g., upcoming speed limit) must be considered. The velocity with respect to vehicles inside the field of view is considered in the safe distance traffic rule concretization.
LTL	$G(\text{keeps_lane_speed_limit} \wedge \text{keeps_fov_speed_limit} \wedge \text{keeps_braking_speed_limit} \wedge \text{keeps_road_condition_speed_limit})$
MTL	$G(\text{keeps_lane_speed_limit} \wedge \text{keeps_fov_speed_limit} \wedge \text{keeps_braking_speed_limit} \wedge \text{keeps_road_condition_speed_limit})$
STL	$G(v_{\text{ego}} \leq v_{\text{sl}}^{\text{max}} \wedge v_{\text{ego}} \leq v_{\text{fov}} \wedge v_{\text{ego}} \leq v_{\text{br}} \wedge v_{\text{ego}} \leq v_{\text{w}})$

Minimum Speed Limit	
StVO §3 (2)	Ohne triftigen Grund dürfen Kraftfahrzeuge nicht so langsam fahren, dass sie den Verkehrsfluss behindern. No motor vehicle must, without good reason, travel so slowly as to impede the flow of traffic.
sign 275	Wer ein Fahrzeug führt, darf nicht langsamer als mit der angegebenen Mindestgeschwindigkeit fahren, sofern nicht Straßen-, Verkehrs-, Sicht- oder Wetterverhältnisse dazu verpflichten. Es verbietet, mit Fahrzeugen, die nicht so schnell fahren können oder dürfen, einen so gekennzeichneten Fahrstreifen zu benutzen. A person operating a vehicle must not do so below the indicated minimum speed unless road, traffic, visibility or weather conditions make this necessary. This sign prohibits vehicles that are unable or not permitted to travel as fast from using a lane marked in this manner.
VCoRT §13 (2)	Similar to maximum speed limit.
Concretization	A vehicle shall consider the minimum speed limit based on a corresponding traffic sign and should not travel so slowly to impede traffic flow.
LTL	$G(\text{keeps_min_speed_limit} \wedge \text{preserves_traffic_flow})$
MTL	$G(\text{keeps_min_speed_limit} \wedge \text{preserves_traffic_flow})$
STL	$G()$

4.3 Highway Traffic Rules

Driving right faster than left

StVO §7 (2), §7 (2a),
§7a Ist der Verkehr so dicht, dass sich auf den Fahrstreifen für eine Richtung
Fahrzeugschlangen gebildet haben, darf rechts schneller als links gefahren werden.

Wenn auf der Fahrbahn für eine Richtung eine Fahrzeugschlange auf dem jeweils linken
Fahrstreifen steht oder langsam fährt, dürfen Fahrzeuge diese mit geringfügig höherer
Geschwindigkeit und mit äußerster Vorsicht rechts überholen.

Auf Ausfädelungstreifen darf nicht schneller gefahren werden als auf den
durchgehenden Fahrstreifen. Stockt oder steht der Verkehr auf den durchgehenden
Fahrstreifen, darf auf dem Ausfädelungstreifen mit mäßiger Geschwindigkeit
und besonderer Vorsicht überholt werden.

Gehen Fahrstreifen, insbesondere auf Autobahnen und Kraftfahrstraßen,
von der durchgehenden Fahrbahn ab, darf beim Abbiegen vom Beginn einer breiten
Leitlinie (Zeichen 340) rechts von dieser schneller als auf der durchgehenden Fahrbahn
gefahren werden.

Auf Ausfädelungstreifen darf nicht schneller gefahren werden als auf den durchgehenden
Fahrstreifen. Stockt oder steht der Verkehr auf den durchgehenden Fahrstreifen, darf
auf dem Ausfädelungstreifen mit mäßiger Geschwindigkeit und besonderer Vorsicht
überholt werden.

StVO English If the density of traffic has resulted in queues of vehicles in the lanes of one
carriageway, traffic on the right (nearside lane, middle lane) may move faster than
traffic on the left (offside lane, middle lane).

If, on a carriageway for one direction of traffic, a vehicle queue has formed and come to
a standstill or is moving at low speed in the left-hand lane, vehicles may overtake this
queue on the right at a slightly higher speed and with the utmost care.

In deceleration lanes, vehicles must not travel faster than traffic on the main carriageway.
If traffic on the main carriageway is moving slowly or is stationary, vehicles in a
deceleration lane may overtake at a moderate speed and with great care.

Where lanes diverge from the main carriageway, especially on motorways and motor roads,
vehicles turning off may, from the beginning of and to the right of a lane marking with
broad lines (sign 340), travel faster than traffic on the main carriageway.

On motorways and other roads outside built-up areas, vehicles may travel faster in
acceleration lanes than traffic on the main carriageway.

Driving right faster than left	
VCoRT	
Concretization	A vehicle shall not drive faster than any vehicle on its left lanes. A exception is, if 1. the vehicle on the left lane is in a congestion; 2. the vehicle on the left lane is on the main carriage way and the vehicle on the right lane is on a off ramp; 3. the left vehicle and right vehicle are separated by a broad lane marking; 4. the right vehicle is within an access ramp. For 1. and 2., the vehicle on the right lane is allowed to drive faster than the vehicle on the left lane with only a slightly higher speed.
Predicates	drives_faster_than_vehicle_left, congestion_left, other_vehicle_left_of_broad_lane_marking, drives_max_with_slightly_higher_speed, right_of_broad_lane_marking
Comments	Assumption: Deceleration lane always right of main carriage way. Case two can be neglected -> same as case one
LTL	$G(\neg \text{drives_faster_than_vehicle_left}$ $\vee (\text{congestion_left} \wedge \text{drives_max_with_slightly_higher_speed})$ $\vee (\text{right_of_broad_lane_marking} \wedge \text{other_vehicle_left_of_broad_lane_marking})$ $\vee (\text{on_access_ramp} \wedge \neg \text{congestion_left}))$
MTL	$G(\neg \text{drives_faster_than_vehicle_left}$ $\vee (\text{congestion_left} \wedge \text{drives_max_with_slightly_higher_speed}))$
STL	

Emergency Lane	
StVO §3 (2)	Sobald Fahrzeuge auf Autobahnen sowie auf Außerortsstraßen mit mindestens zwei Fahrstreifen für eine Richtung mit Schrittgeschwindigkeit fahren oder sich die Fahrzeuge im Stillstand befinden, müssen diese Fahrzeuge für die Durchfahrt von Polizei- und Hilfsfahrzeugen zwischen dem äußerst linken und dem unmittelbar rechts daneben liegenden Fahrstreifen für eine Richtung eine freie Gasse bilden.
StVO English	As soon as vehicles on motorways and roads outside a built-up area with at least two lanes for one direction start to move at walking pace or come to a standstill, these vehicles must leave a gap for one direction between the lane on the far left and the lane immediately adjacent to it on the right to allow police and emergency vehicles to pass.
VCoRT	
Concretization	A vehicle in a congestion outside urban roads has to drive leftmost if it is in the leftmost lane and has to drive rightmost otherwise.
Predicates	in_left_most_lane, drives_left_most
Comments	
LTL	$G((\text{in_left_most_lane} \implies \text{drives_left_most})$ $\wedge (\neg \text{in_left_most_lane} \implies \text{drives_right_most}))$
MTL	$G((\text{in_left_most_lane} \implies \text{drives_left_most})$ $\wedge (\neg \text{in_left_most_lane} \implies \text{drives_right_most}))$
STL	

No reversing and turning	
StVO §18 (7)	Wenden und Rückwärtsfahren sind verboten.
StVO English	Making U-turns and reversing are prohibited.
VCoRT	
Concretization	Making U-turns and reversing are prohibited.
Predicates	makes_u_turn, reverses
Comments	
LTL	$G(\neg \text{makes_u_turn} \wedge \neg \text{reverses})$
MTL	$G(\neg \text{makes_u_turn} \wedge \neg \text{reverses})$
STL	

Direction indicator

StVO §5 (4a)	Das Ausscheren zum Überholen und das Wiedereinordnen sind rechtzeitig und deutlich anzukündigen; dabei sind die Fahrtrichtungsanzeiger zu benutzen. The intention of pulling out to overtake and of moving back to the right-hand side of the road are to be signalled clearly and in good time, using the vehicle's direction indicators to do so.
StVO §7 (5)	[...] Jeder Fahrstreifenwechsel ist rechtzeitig und deutlich anzukündigen; dabei sind die Fahrtrichtungsanzeiger zu benutzen. [...] Any change of lane must be signalized clearly and in good time, using the vehicle's direction indicators to do so.
Burmann, et al., Straßenverkehrsrecht, Recital 11	Der Fahrtrichtungsanzeiger ist rechtzeitig bevor Verlassen der Hauptfahrbahn und dem Befahren des Ausfädelungsstreifen zu benutzen. Sobald der Ausfädelungsstreifen befahren wird, soll der Fahrtrichtungsanzeiger deaktiviert werden. The direction indicator shall be timely switched on before leaving the main carriageway and moving to the exit ramp. When the exit ramp had been entered, the direction indicator should be switched off.
VCoRT	
Concretization	Any change of lane must be signalized clearly and enough time before entering the new lane, using the vehicle's direction indicators. When the new lane had been entered, the direction indicator should be switched off.
Predicates	
Comments	We assume the exit ramp is long enough so that a vehicle can switch off the direction indicator before
LTL	$G()$
MTL	$G()$
STL	

4.4 Urban Traffic Rules

4.5 Cross-Country Road Traffic Rules

References

- [1] Economic Comission for Europe: Inland Transport Committee, “Vienna Convention on Road Traffic,” Nov. 1968.
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- [3] C. Pek and M. Althoff, “Efficient Computation of Invariably Safe States for Motion Planning of Self-Driving Vehicles,” in *Proc. of the IEEE Int. Conf. on Intelligent Robots and Systems*, 2018, pp. 3523–3530.